TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test (CT)-2
Semester: V
Programme: B. Tech Information Technology
Course: Theory of Computation

Time: 1 Hr 60 min **Max Marks:** 30 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33502	Course Outcome
CO1	To illustrate finite state machines to solve problems in computing.
CO2	To classify theoretical foundations of computer science from the perspective of formal languages.
CO3	To familiarize Regular grammars and Context Free Grammars.
CO4	To enhance students' ability to understand and conduct mathematical proofs for computation and algorithms.
CO5	To explain hierarchy of problems arising in computer science.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define ambiguity in CFG.	CO3	2	2
b.	Define TM.	CO4	2	2
c.	Define NP problem.	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain ambiguous grammar with example.	CO3	2	4

b.	Explain CNF and GNF.	CO3	2	4
c.	Explain PDA acceptance.	C03	2	4
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain TM design.	CO4	2	4
b.	Explain non-deterministic TM.	CO4	3	4
c.	Explain recursive languages.	CO4	3	4
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain decidability.	CO5	2	4
b.	Explain Church's hypothesis.	CO5	3	4
c.	Explain Ackermann function.	CO5	4	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create